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DexForesight

Grounding Dexterous Policies with Implicit Spatial Foresight

Implicit spatial foresight: learned in training, never generated at test

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Project page: dexforesight.github.io

TL;DR

Distill world-model spatial foresight into policy latents

Training: an action-conditioned spatial world model, trained with offline depth, kinematic hand point clouds and demo actions, predicts 3D hand point flow and DINOv3 feature changes; frozen, its features are injected into the VLA policy via query alignment. **Deployment:** only RGB + language + robot state; no world model, depth estimation or future decoding.

49.4% → 60.5%

Overall success, 11 DexJoCo sim tasks (GR00T N1.7 base, +11.1 pts)

58.3% vs 50.0%

Overall success, 3 real dexterous tasks (vs. strongest baseline WLA)

+3.0 ms

Latency 227.4 → 230.4 ms (+1.3%) vs. QwenVLA base, same hardware

Core claim: predictive knowledge need not be generated at deployment; the policy can absorb it as a training-time privileged signal.

Dexterity needs foresight; prior routes cost at deployment

Explicit future generation

World-action models (WAMs)

- Generate future images/motion at inference, then act
- Extra compute every step: Cosmos3 needs 930 ms for 16 video frames
- Must reconstruct action-irrelevant appearance (texture, background)

Depth input at deployment

3D / depth-conditioned policies

- Rely on an online depth reconstruction pipeline
- Extra compute; needs temporally consistent depth video as hand and objects move

Static spatial supervision

ReconVLA, Spatial Forcing, etc.

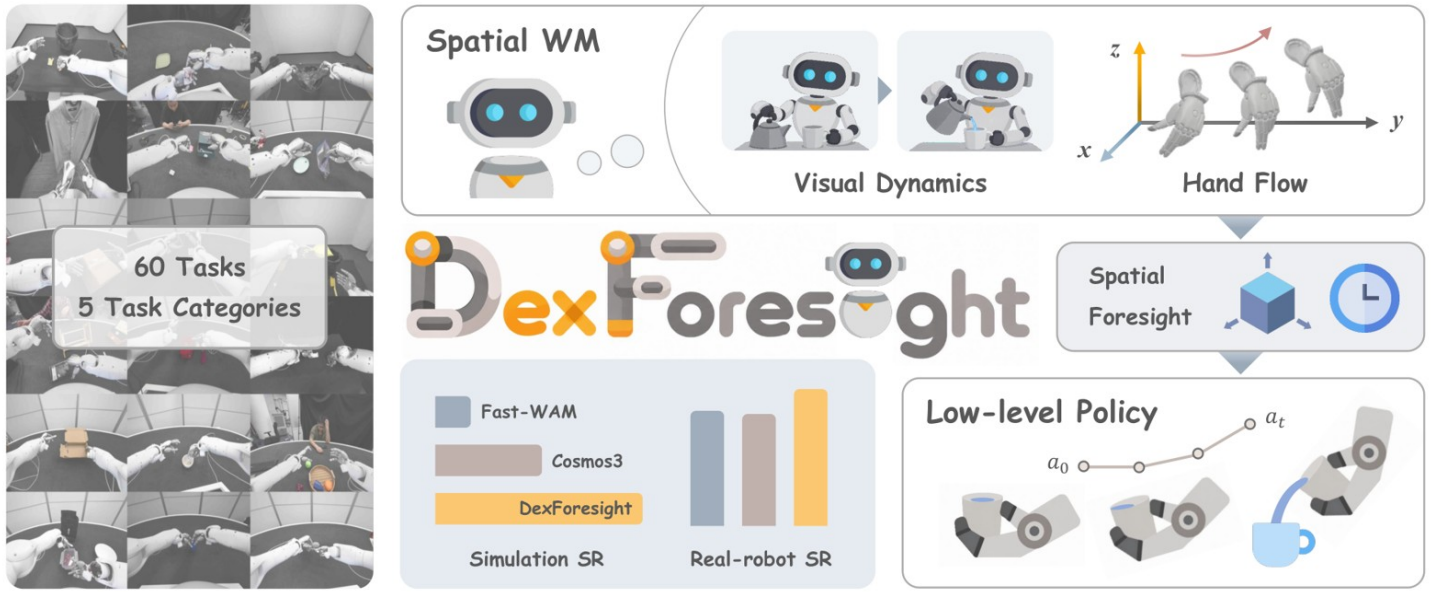
- Mainly improve scene-geometry representations
- Do not model how the controlled multi-finger hand moves

Shared blind spot: cameras miss occluded fingers; kinematics + joint states locate all hand links, visible or not.

Research question: can training learn predictive features for spatial foresight, decoding no future at deployment?

Source: Secs. 1, 2; Sec. 4.2 (Cosmos3 latency)

World model: training-only spatial teacher; deploy policy alone



Training only (privileged info)

- Multi-view RGB, language, robot state
- Offline depth (real: StereoAnyVideo / sim: rendered)
- Hand point clouds from URDF + forward kinematics
- Demo actions and recorded future states

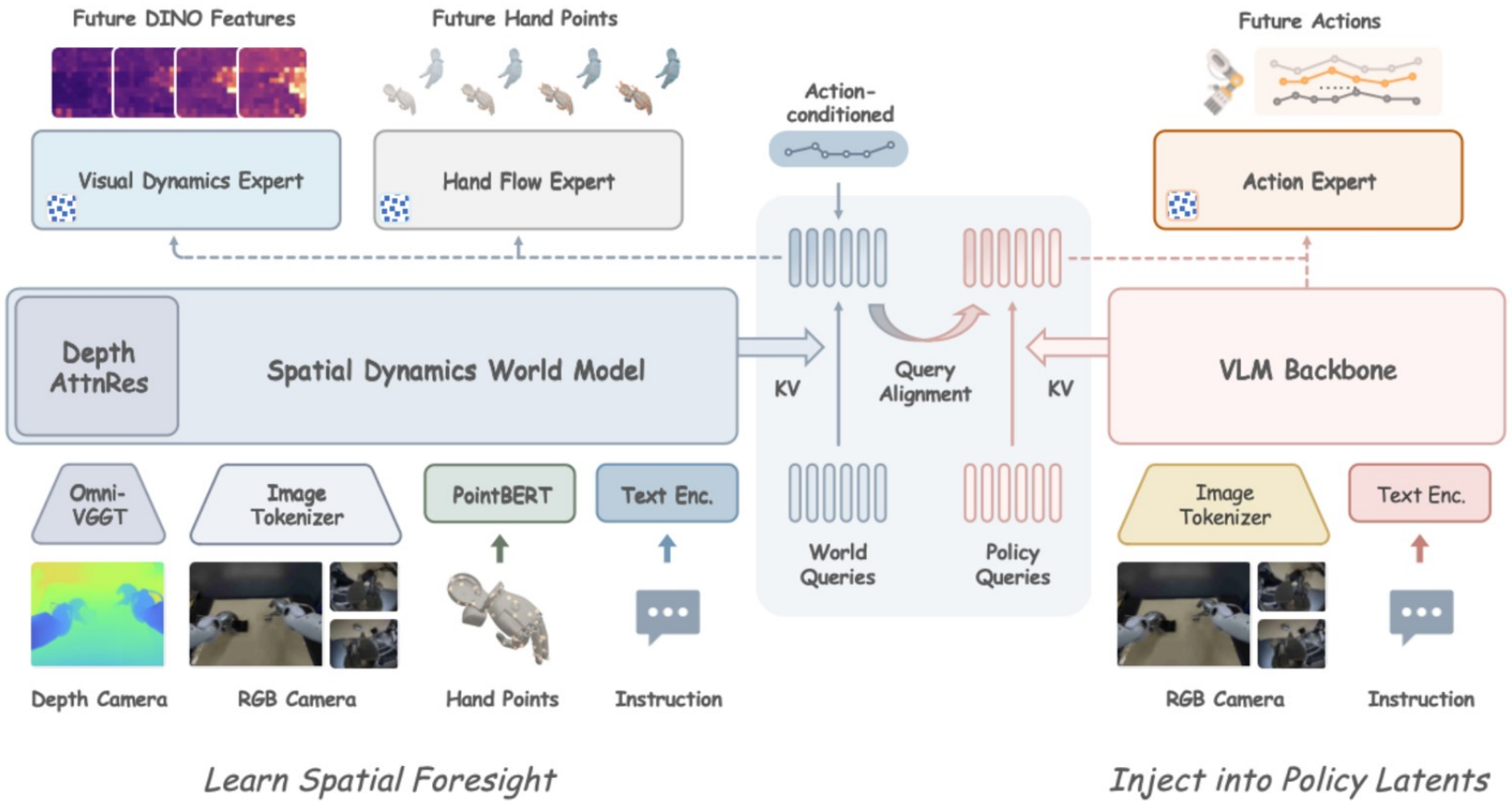
▼ Query-alignment distillation

Deployment uses only

- Current RGB + language + robot state
- No world model, no online depth estimation
- No future images / motion generated

Source: Figure 1; Secs. 1, 3

2 stages: action-cond. WM → query align. → action expert



1 Spatial-dynamics WM

Jointly encodes RGB, offline depth (OmniVGGT + Depth AttnRes), hand points (Point-BERT) and instruction; backbone initialized from RynnBrain

2 Action-cond. world queries

Demo actions modulate world queries via a query adapter, driving Hand Flow / Visual Dynamics experts to predict the future

3 Query align → act. expert

Teacher frozen; policy-VLM policy queries align to world queries and join the Action Expert's context

Teacher: occluded-finger 3D flow + scene feature change

Hand geometry

- Fixed points sampled on URDF links, mapped to base frame via FK on recorded joint states
- Each point keeps its link and local coords → fixed cross-frame correspondence
- Occluded fingers still have exact coordinates
- Point-BERT encodes current cloud + short history

Depth fusion

- OmniVGGT encodes RGB-D into geometry tokens
- Injected via cross-attention after layers 2, 6, …, 26 (learnable scale γ)
- Token-wise softmax routing (Attention Residuals) before layers 4, 8, …, 28
- Initialized near pretrained compute, preserving features

Action-cond. prediction

- N world queries read layers 12/20/28
- 16-frame future window in 4 bins, 1 offset per bin (K=4)
- Demo actions (relative to current state) via adapter
- Two diffusion experts + Smooth L1; future-horizon dropout

Hand Flow: $F = P(t+\delta) - P(t)$, 3D displacement of M hand anchors

Visual Dynamics: $\Delta V = \Phi(I(t+\delta)) - \Phi(I(t))$, DINOv3 feature residual

$$L_{world} = L_{hand} + \lambda_v \cdot L_{visual} + \lambda_c \cdot L_{context}$$

Depth is only a current geometric condition; no future depth targets. Actions are conditioning inputs; hand motion and visual change are the targets.

Stage 2 • Injection into Policy Latent Space

How the student absorbs it: query alignment + action context, only +3 ms

1 Frozen teacher

Runs only the world-model backbone + query adapter to get Z^w; both predictive experts skipped

2 Align queries

N new policy queries read policy-VLM hidden layers, projected to teacher space: per-query cosine (token) + pairwise similarity matrix (relation)

3 Condition actions

Aligned queries projected by W_{act} and concatenated: $\tilde{C} = [C ; Z^{\pi} \cdot W_{act}]$, trained jointly with the action loss

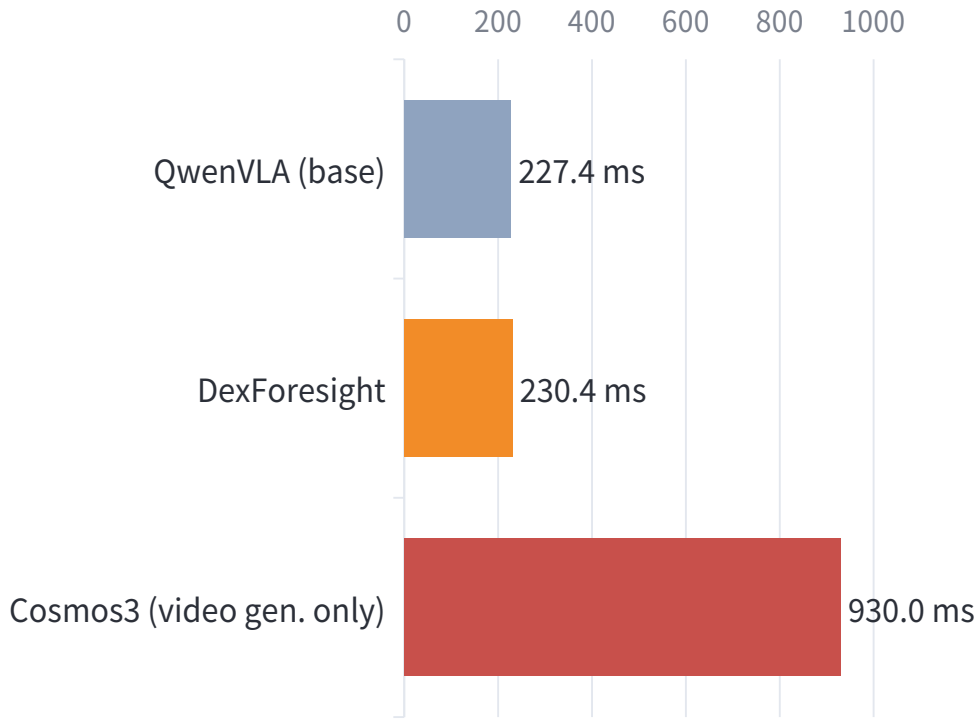
4 Deploy

World model, depth/hand encoders and both experts removed; context computed once per observation, reused in sampling

$$L_{policy} = L_{action} + \lambda_{align} (L_{token} + \beta \cdot L_{relation})$$

Note: the student never sees demo actions U_t; alignment is an auxiliary objective, not exact reproduction of teacher predictions

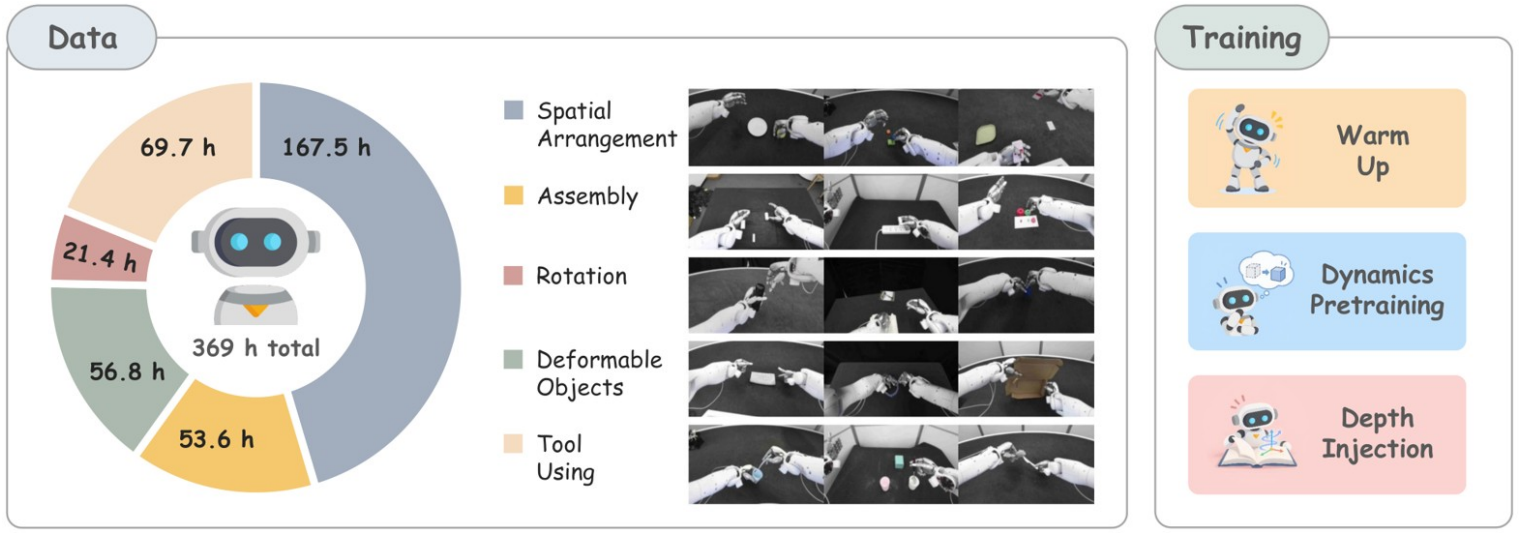
Per-inference latency (ms, lower is better)



Same hardware/settings: DexForesight adds only 3.0 ms over QwenVLA (+1.3%). Cosmos3's 930 ms covers only its 16-frame future video generation.

Source: Sec. 3.2, Eq. (5); Sec. 4.2 (latency)

WM trained in 3 stages on ~370 h real demos, 60 tasks



369 h

60 tasks • 5 categories (Figure 3 totals)

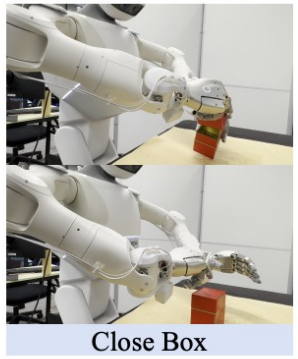
Category	Hours
Spatial placement	167.5 h
Tool use	69.7 h
Deformables	56.8 h
Assembly & contact	53.6 h
Rotation & articulation	21.4 h

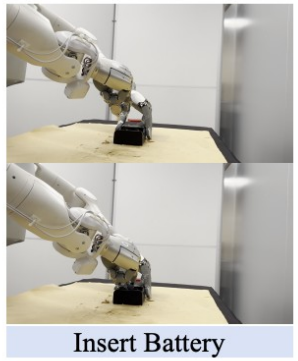
Three stages: Warmup → dynamics pretraining (no depth) → depth injection

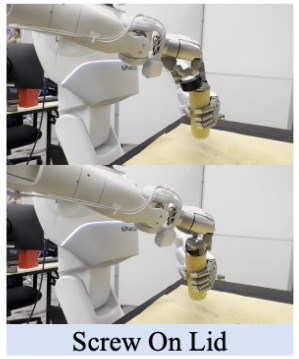
Depth stage: 100 h subset covering placement / assembly / rotation; depth estimated offline by StereoAnyVideo

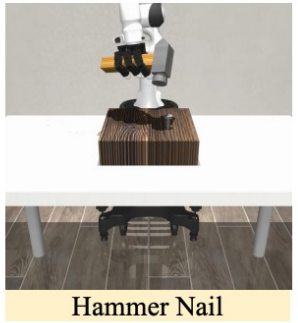
Source: Figure 3; Sec. 3.1 Data and training

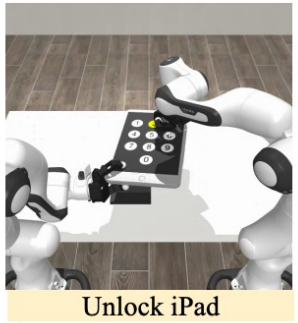
Evaluation: 11 simulated tasks and 3 challenging real dexterous tasks

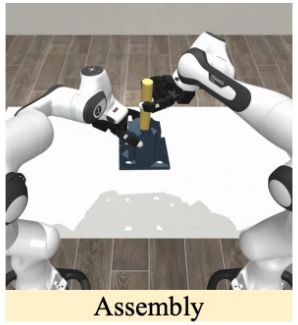

Close Box


Insert Battery


Screw On Lid


Hammer Nail


Unlock iPad


Assembly

Sim • DexJoCo

- 11 tasks, single- + bimanual
- Franka Panda + 16-DoF Allegro hand
- GR00T N1.7 base; depth rendered by MuJoCo
- 100 trials per task

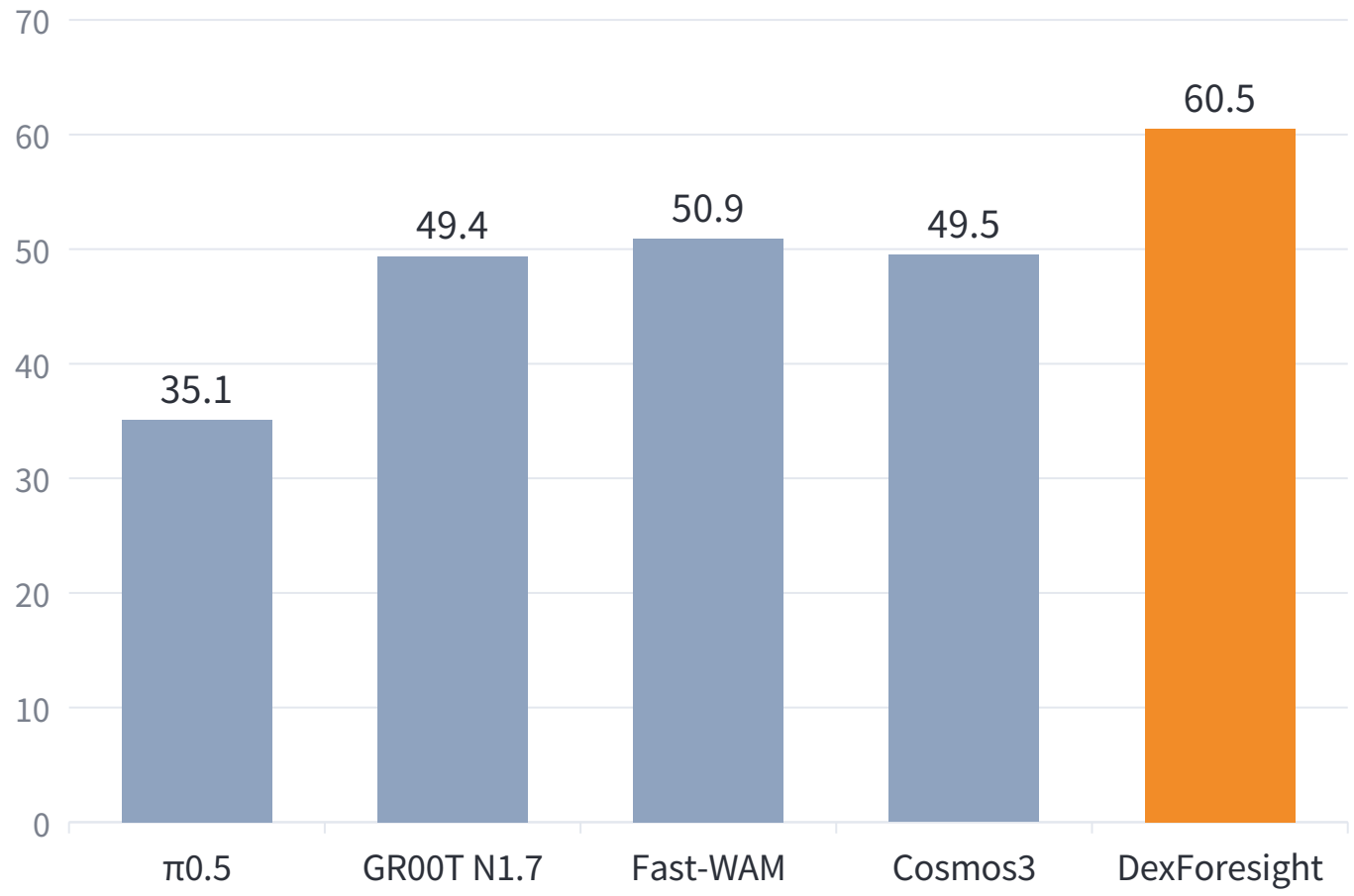
Real • Sharpa North

- 22-DoF Sharpa Wave hand; QwenVLA base
- Close Box / Insert Battery / Screw On Lid
- Test positions unseen in training; Screw also changes cup color
- All methods fine-tuned 30 epochs; 20 trials per task

Source: Figure 4; Sec. 4.1

Sim: 60.5% overall, +9.6 pts over best baseline Fast-WAM

DexJoCo overall success across 11 tasks (%)



+11.1 pts

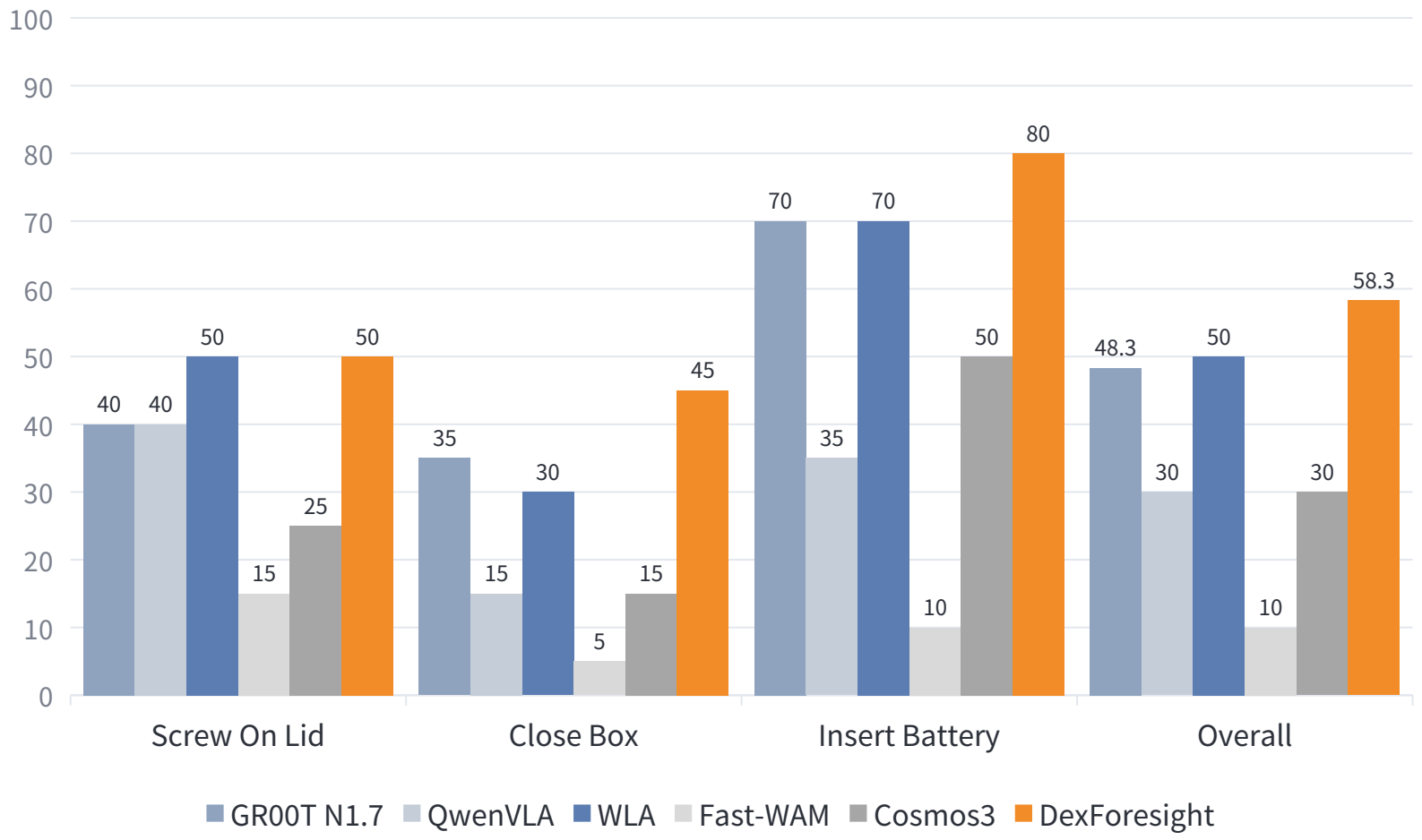
Over its own GR00T N1.7 base (49.4 → 60.5)

Method	Single-arm	Bimanual
π0.5	53.0	13.6
GR00T N1.7	61.5	34.8
Fast-WAM	60.2	39.8
Cosmos3	59.5	37.4
DexForesight	**76.3**	**41.6**

Largest gain single-arm (+14.8 vs. GR00T); bimanual only 1.8 pts above Fast-WAM. 100 trials per task.

Real: 58.3% overall, best on all three tasks (tied with WLA on Screw)

Real-robot success (%, 20 trials/task, test positions unseen in training)



+28.3 pts

Over QwenVLA base (30.0 → 58.3)

35/60 vs 30/60

Lead over WLA is few trials; small sample

Failure modes match design challenges

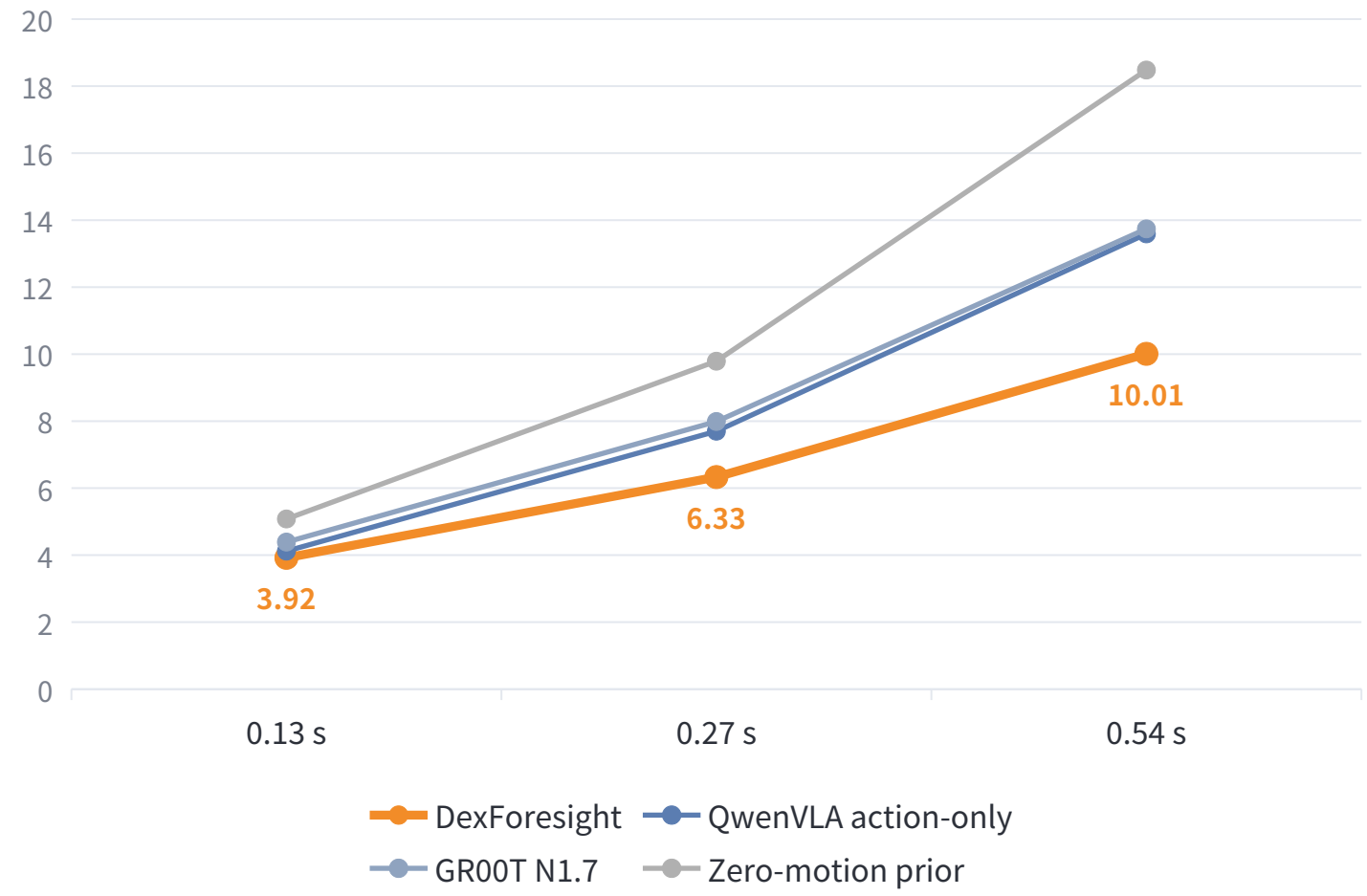
Screw: finger occlusion

Insert: partial object occlusion

Close: lid–box geometry

Teacher-free policy features retain more future hand motion

Linear-probe error, future 3D hand-point shift (mm, ↓)



−26.3%

Error at 0.54 s: 13.59 → 10.01 mm
(vs. same-architecture QwenVLA action-only)

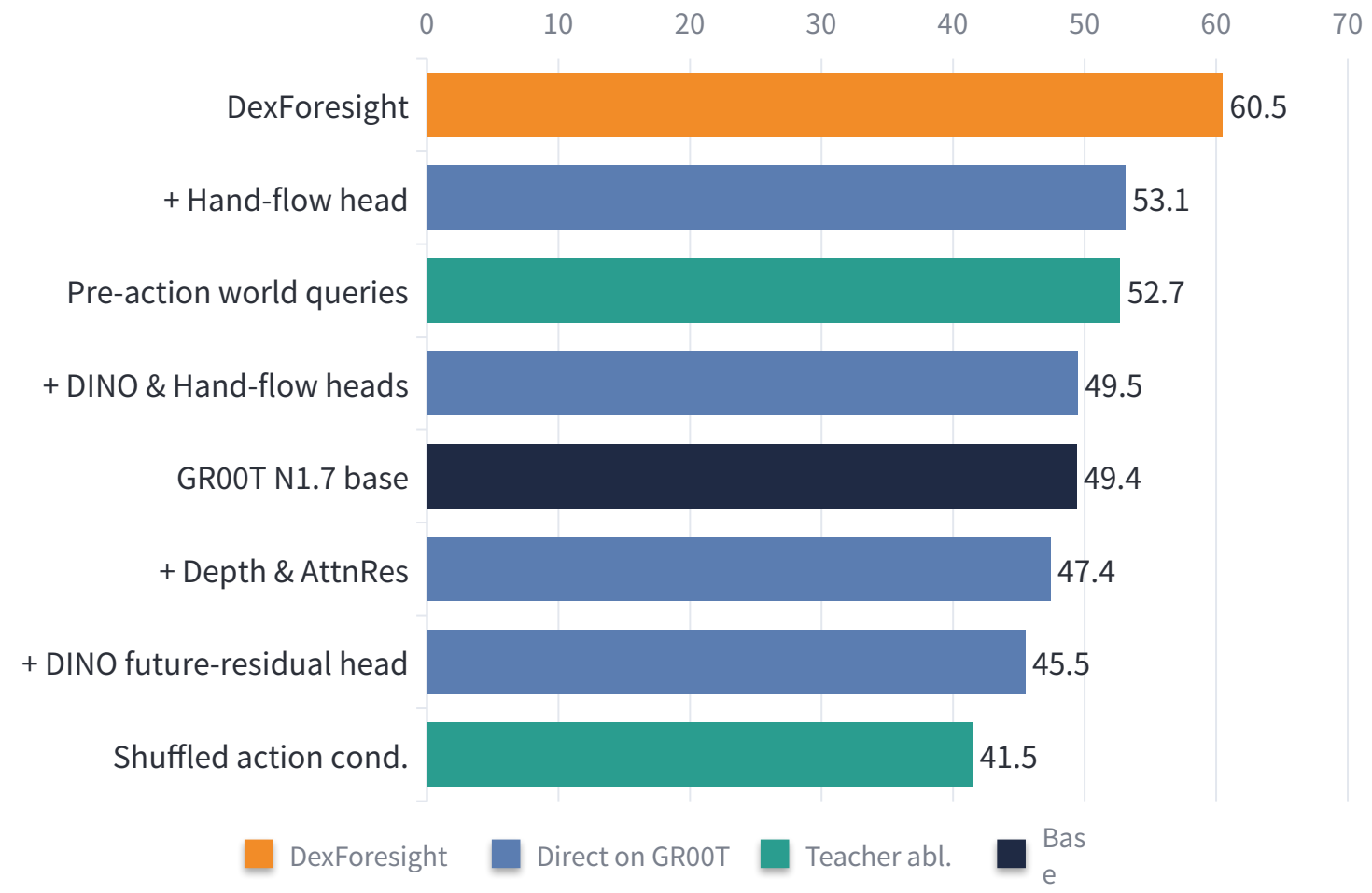
Probe protocol

- Frozen real-robot checkpoints; features from the policy only
- Ridge-regression linear probe; episode-level train/test split
- 95% episode-bootstrap confidence intervals

Caveat: the gap grows with horizon, but CIs overlap at 0.13 s ([3.47, 4.13] vs. [3.83, 4.47]); short-horizon difference not significant.

Distillation beats direct heads; teacher needs true actions

DexJoCo overall success (%)



Direct heads trail distillation

Best direct option, Hand-flow head, gets 53.1%; DexForesight is 7.4 pts higher; both heads together drop to 49.5%

Action conditioning is key

Non-action-conditioned pre-action queries reach only 52.7%; conditioning on true actions lifts it to 60.5%

A wrong teacher hurts the policy

Shuffled action conditioning drops to 41.5%, 7.9 pts below the base

Source: Table 4 (top, middle groups); Sec. 4.3

Teacher quality matters: depth, fusion, hand flow all needed

No offline depth

49.7%

Sim −10.8 pts

Real 58.3 → 43.3 (−15.0)
Screw 35 / Close 25 / Insert 70
Deployment inputs unchanged

Concat depth tokens instead of cross-layer fusion

42.0%

Sim −18.5 pts

Authors' conjecture: concatenated depth tokens
perturb pretrained multimodal features

Future-action targets instead of hand point flow

48.4%

Sim −12.1 pts

Below the 49.4% GR00T base
Supports kinematic hand-flow supervision
(covers occluded hand surfaces)

Reference: full DexForesight 60.5% sim / 58.3% real; GR00T N1.7 base 49.4% sim. Each variant changes only the teacher; student inputs unchanged.

Critical Assessment

Clean idea, full evidence; real stats, baselines need work

Strengths

Privileged-info distillation: depth, point clouds, future states used only in training; +3 ms at deployment

Kinematic hand point flow: immune to occlusion, targeting finger occlusion in dexterous manipulation

Complete evidence chain: success rates + linear probes + direct-head comparison answer why to train a teacher, then distill

Useful negative results: direct heads and concatenated depth tokens both hurt

Limitations & Questions

Small real-robot sample: 20 trials/task; 35/60 vs. 30/60 against WLA; tied on Screw

Questionable baseline adaptation: Fast-WAM gets 50.9% in sim but only 10% real

Sim–real inconsistency: without depth, sim ≈ base, yet real stays clearly above base

Negative-transfer risk: shuffled actions fall below base; the student sees no actions, so alignment targets may average out under multimodal behavior (not analyzed)

Cost & generalization: ~370 h data + a separately trained large world model; only 3 real tasks, generalization tested only on new positions and colors

TAKEAWAYS

Policy foresight, no world model at deployment

01	**Training-time privileged signals**	Offline depth + kinematic hand point clouds + demo actions train an action-conditioned spatial world model
02	**Choice of prediction targets**	3D hand point flow + DINOv3 feature residuals; future-action targets or concatenated depth tokens perform clearly worse
03	**Query-alignment injection**	Aligned queries join the action context directly; sim +11.1 pts, real 58.3%, latency +3 ms

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